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using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace WindowsFormsApplication13
{
    public partial class Form1 : Form
    {
        int[] a = new int[16];
        int[] b = new int[16];
        int[] c = new int[16];
        Button[] kutu = new Button[16];
        int enbsira;

        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            listBox1.Items.Clear();
            for (int i = 0; i < 16; i++)
            {
                Random rastgele = new Random();
                a[i] = rastgele.Next(100000);
                b[i] = a[i];
                listBox1.Items.Add(a[i]);
                System.Threading.Thread.Sleep(500);
            }
        }

        private void button3_Click(object sender, EventArgs e)
        {
            for (int i = 0; i < 16; i++)
            {
                c[i] = Array.IndexOf(a, b[i])+1;
                listBox3.Items.Add(c[i]);
                kutu[i].Text = c[i].ToString();
            }
            enbsira = Array.IndexOf(c, c.Max());
            kutu[enbsira].Text = "";
            kutu[enbsira].BackColor = this.BackColor;
        }

        private void button2_Click(object sender, EventArgs e)
        {
            Array.Sort(b);
            listBox2.Items.Clear();
            for (int i = 0; i < 16; i++)

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        listBox2.Items.Add(b[i]);
    }

    private void checkBox1_CheckedChanged(object sender, EventArgs e)
    {
        groupBox1.Visible = checkBox1.Checked;
    }

    private void puzzle(object sender, EventArgs e)
    {
        Button tiklanan = sender as Button;
        int farkx = tiklanan.Left - kutu[enbsira].Left;
        int farky = tiklanan.Top - kutu[enbsira].Top;
        int toplam = Math.Abs(farkx) + Math.Abs(farky);
        if (toplam < 111)
        {
            int tx = tiklanan.Left;
            int ty = tiklanan.Top;

            tiklanan.Left = kutu[enbsira].Left;
            tiklanan.Top = kutu[enbsira].Top;

            kutu[enbsira].Top = ty;
            kutu[enbsira].Left = tx;
        }
    }

    private void Form1_Load(object sender, EventArgs e)
    {
        int sayac = 0;

        for (int satir = 0; satir < 4; satir++)
            for (int sutun = 0; sutun < 4; sutun++)
            {
                kutu[sayac] = new Button();
                kutu[sayac].Width = ornek.Width;
                kutu[sayac].Height = ornek.Height;
                kutu[sayac].Font = ornek.Font;
                kutu[sayac].BackColor = ornek.BackColor;
                kutu[sayac].ForeColor = ornek.ForeColor;
                kutu[sayac].Text = "";
                kutu[sayac].TextAlign = ornek.TextAlign;
                kutu[sayac].Left = sutun * ornek.Width;
                kutu[sayac].Top = satir * ornek.Height;
                kutu[sayac].Click += new System.EventHandler(puzzle);
                this.Controls.Add(kutu[sayac]);
                sayac++;
            }
    }
}

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